

Advanced Task 4-1 HC-05 Master/Slave Communication

Work in pairs — you'll also practice GitHub project collaboration



Student A

HC-05 = MASTER

- Configure HC-05 as MASTER
- Circuit: push button + DC motor



Student B

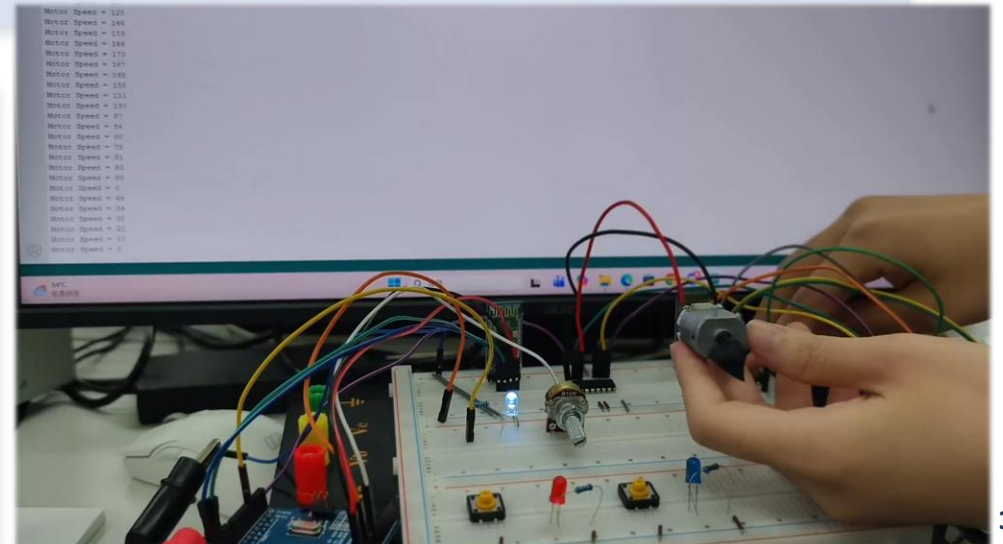
HC-05 = SLAVE

- Configure HC-05 as SLAVE
- Circuit: variable resistor + LED



Bidirectional control protocol

- A presses/releases the button → B's LED turns on/off
- B rotates the potentiometer → A's motor spins at the matching speed



Advanced Task 4-1 GitHub Collaboration

Objective: practice Git workflow, repository management, role-based branch control, and Code Review with GitHub's Pull Request feature.

Repository structure

```
Bluetooth-Bidirectional-Control/      (Repository Root)
├── README.md                        (Maintained by A & B together ; Explanation of the task)
├── Master_Node/
│   └── Master_Node.ino              (Developed by Student A)
└── Slave_Node/
    └── Slave_Node.ino               (Developed by Student B)
```

Advanced Task 4-1 Git Workflow, Step by Step

Student A — Project Manager

1. Create Repository

Create a new Public repo named “Bluetooth-Bidirectional-Control”

2. Initialize Structure

Add README.md (title + hardware description) and two folders: /Master_Node and /Slave_Node

3. Invite Collaborator

Settings → Collaborators → invite Student B

4. Develop on Your Own Branch

Like Student B, work on your own feature branch instead of developing directly on main

5. Manage the Merge

As project manager, merge both branches into main

Student B — Collaborator

1. Clone Repository

```
git clone <Repo_URL>
```

2. Create Feature Branch

Must NOT push directly to main:

```
git checkout -b feature/slave-firmware
```

3. Develop & Commit

Write Slave_Node.ino inside /Slave_Node/, then:

```
git add Slave_Node/Slave_Node.ino
git commit -m "Add Slave node code for Potentiometer and LED control"
git push origin feature/slave-firmware
```